

AI research card

Keep this beside you when you use AI on a research question. It keeps the judgment with you.

YOUR QUESTION

YOUR SENSE OF RESEARCH AND ITS ROUTE

Circle the phases your question will pass through, then note what skipping the others risks.



THREE HABITS THAT PROTECT YOUR JUDGMENT

Generate first

Do: Write a three-sentence answer or a prediction first. Then ask the AI what you missed.

Ask the AI: “Here is my draft answer. What did I miss, and which of my claims is weakest?”

Why: In a field trial, unrestricted AI raised practice performance and lowered unassisted performance; guardrails removed the harm. (Bastani et al. 2025)

Read the source yourself

Do: Before you accept a synthesis, open the top source and read the parts it is cited for.

Ask the AI: “For each claim, quote the sentence in the source that supports it and say where it appears.”

Why: Most tools read only titles and abstracts. In 6,063 claims from abstracts on AI and learning, 82.1% of harm claims reported no statistical test, against 70.8% of benefit claims. (V. N. Patel 2026)

Check every citation

Do: Look up every citation, and open any you plan to repeat.

Ask the AI: “List each source with its DOI or link, so I can check it.”

Why: Published tests found 55% of GPT-3.5 citations and 18% of GPT-4 citations fabricated. (Walters & Wilder 2023)

THE EIGHT HABITS: TICK THE ONES YOU USED LAST TIME

- ☐ **Frame the question first.** Write the question and what would count as an answer before you open the tool.
- ☐ **Read the source yourself.** Before you accept a synthesis, open the top source and read the parts it is cited for.
- ☐ **Use more than one door.** Run the same question in at least two tools, and compare what each one found.
- ☐ **Set a budget and a stop rule.** Before you start, set the time the question is worth and a rule for stopping. Extend it only by a deliberate decision.
- ☐ **Generate first.** Write a three-sentence answer or a prediction first. Then ask the AI what you missed.
- ☐ **Check every citation.** Look up every citation, and open any you plan to repeat.
- ☐ **Look for evidence against.** Keep an evidence-against column, and ask for the strongest case against your view.
- ☐ **Plan when to look again.** Note when you will revisit the answer, and what new evidence would change it.

TAKE-HOME MESSAGES

- Say which “research” you mean. The sense sets the route, and the route shows which phases you skip and what that risks.
- Generate first, then check: write your frame, prediction or draft before you open the tool.
- Different tools find different papers. For the same question, the four tools’ top 20 results overlapped by about 1%. Search more than one, and read the sources yourself.
- Check every citation. Published tests found 18% to 55% of AI citations fabricated.
- A loop is only virtuous if something in it can say no: data, a dissenter, a prediction written beforehand, or a source you opened yourself.
- Set a time budget and a stop rule before you start.

Sources. Bastani, H. et al. Generative AI without guardrails can harm learning: evidence from high school mathematics. *Proc. Natl Acad. Sci. USA* 122, e2422633122 (2025), doi:10.1073/pnas.2422633122. · Patel, V. N. Under-reported harm: a claim-level audit of abstract reporting on AI assistance and human learning. In *Human-AI Coevolution: Measuring Human-Agent Teams in the Agentic Era* (HAIC workshop, NeurIPS 2026; non-archival). <https://github.com/vpatel43/Effect-of-AI-Assistance-on-Learning-Claim-level-evidence-audit> (2026). · Walters, W. H. & Wilder, E. I. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Sci. Rep.* 13, 14045 (2023), doi:10.1038/s41598-023-41032-5.

Make your own card: answer the three questions in “What do you mean, research?”, then choose “Make my card”.
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Interactive Case Studies